Exercise 1: Rewind – CNN and RNN

Task 1: CNN Basics

- Why is a CNN generally better suited for image processing compared to an MLP?
- Explain why CNN architectures usually consist of *multiple* convolutional layers.
- Describe the purpose of pooling operations in CNNs. Discuss what effects it would have if pooling was completely omitted.

Task 2: RNN Basics

- What is the main difference between feedforward neural networks and recurrent neural networks regarding their network structure?
- What role does the hidden state play in an RNN, and how does it enable sequential data processing?
- What is the vanishing gradient problem in RNNs, and why does it occur?

Task 3: Classification metrics

A binary classification model predicts whether a data point belongs to the positive (+) or negative (−) class. The table below shows the predicted and expected (actual) labels for 15 data points.

Data point	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**
Predicted	+	−	−	+	−	−	+	−	+	−	−	−	+	−	−
Expected	+	−	+	+	−	−	−	+	+	−	+	−	+	+	−

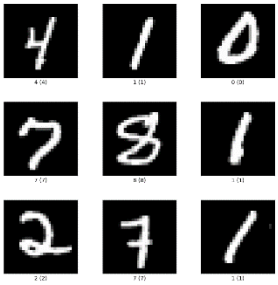
Tasks:

1. Construct the confusion matrix for the classifier. Identify the number of True Positives (TP), False Positives (FP), True Negatives (TN), and False Negatives (FN).
2. Calculate the **precision** of the classifier.
3. Calculate the **recall** of the classifier.

Show your calculations and express both metrics as percentages. Treat + as the positive class.

Task 4: MNIST

Use the file *mnist.ipynb* provided in the Moodle course to implement a CNN for solving the mnist digits dataset (https://www.tensorflow.org/datasets/catalog/mnist).
Each sample has a 28x28 grayscale image of a single handwritten digit (0-9), as shown on the right, and the corresponding expected output, again is a digit from 0 to 9.



Task 5: Thus Spoke Zarathustra

Check out the *zarathustra.ipynb* from Moodle. This notebook will guide you through training an RNN that is able to generate text via predicting the next word from a series of previous words. The model will be trained on the *zarathustra.txt*, a classic work of German literature, which you can also download from Moodle.